

# 1. Autonomous Vehicle Vision System Selection

## Problem Context

An automobile company is developing an autonomous vehicle detection system that must satisfy the following operating constraints:

- Accuracy constraint: if accuracy < 90%, the system fails.
- Latency constraint: if latency > 70 ms, the response is unacceptable.
- Computation constraint: the vehicle platform supports only moderate computational resources.

## Candidate Methods

| Method              | Accuracy | Latency | Computation | Accuracy ≥ 90% | Latency ≤ 70 ms | Computation ≤ Moderate |
|---------------------|----------|---------|-------------|----------------|-----------------|------------------------|
| Viola-Jones         | 82%      | 30 ms   | Low         | No             | Yes             | Yes                    |
| YOLO                | 93%      | 60 ms   | Moderate    | Yes            | Yes             | Yes                    |
| Deep Learning (CNN) | 97%      | 90 ms   | High        | Yes            | No              | No                     |

## Constraint-by-Constraint Analysis

### Viola-Jones:

Viola-Jones offers the lowest latency (30 ms) and the lowest computational load, making it very efficient. However, its accuracy of 82% falls well below the 90% threshold. Since the accuracy constraint is a hard failure condition, Viola-Jones is eliminated regardless of its favorable speed and resource profile. In a safety-critical domain such as autonomous driving, an 18-percentage-point accuracy shortfall translates directly into missed or misclassified obstacles, pedestrians, or signals — an unacceptable risk.

### YOLO:

YOLO achieves 93% accuracy, comfortably clearing the 90% minimum, and its latency of 60 ms stays within the 70 ms ceiling, leaving a 10 ms margin for real-time responsiveness. Its computational demand is described as moderate, which matches the vehicle's available computing resources exactly. YOLO is therefore the only method that satisfies all three constraints simultaneously.

**Deep Learning (generic CNN pipeline):**

This method delivers the highest accuracy at 97%, which would be ideal from a pure detection-quality standpoint. However, its latency of 90 ms exceeds the 70 ms limit by 20 ms, and it requires high computational resources that surpass what the vehicle platform supports. Both violations are independently disqualifying: the latency breach risks delayed hazard response at driving speeds, and the computational mismatch would likely force the system to throttle performance or require hardware the vehicle does not have.

Decision

**YOLO is the method that satisfies all stated constraints and should be selected for the autonomous vehicle vision system.**

The reasoning follows a simple elimination logic: Viola-Jones fails the accuracy gate, Deep Learning fails both the latency and computation gates, and YOLO is the only candidate that clears every gate. This also reflects a well-known real-world trade-off — classical detectors (Viola-Jones) are fast but insufficiently accurate for complex, dynamic driving scenes, while heavy deep networks are accurate but too slow and resource-intensive for real-time, resource-constrained embedded deployment. YOLO's single-pass detection architecture is specifically designed to balance this trade-off, which is exactly why it emerges as the constraint-satisfying choice here.

2. Smart Factory Product Inspection Selection

Problem Context

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line, under the following constraints:

- Accuracy constraint: if accuracy < 95%, defects may be missed.
- Processing time constraint: if processing time > 50 ms, production throughput is affected.
- Computation constraint: the hardware supports only moderate computation.

Candidate Methods

| Method                     | Accuracy | Processing Time | Computation | Accuracy ≥ 95% | Time ≤ 50 ms | Computation ≤ Moderate |
|----------------------------|----------|-----------------|-------------|----------------|--------------|------------------------|
| Traditional Image Analysis | 91%      | 25 ms           | Low         | No             | Yes          | Yes                    |
| YOLO                       | 96%      | 45 ms           | Moderate    | Yes            | Yes          | Yes                    |

| Method              | Accuracy | Processing Time | Computation | Accuracy $\geq$ 95% | Time $\leq$ 50 ms | Computation $\leq$ Moderate |
|---------------------|----------|-----------------|-------------|---------------------|-------------------|-----------------------------|
| Deep Learning (CNN) | 98%      | 75 ms           | High        | Yes                 | No                | No                          |

## Constraint-by-Constraint Analysis

### Traditional Image Analysis:

This approach is extremely fast (25 ms) and computationally light, which suits high-speed production lines well. However, its accuracy of 91% is below the 95% threshold required to reliably catch defects. Because missed defects directly translate into faulty products reaching customers, this accuracy gap is a disqualifying failure regardless of its speed advantage.

### YOLO:

YOLO reaches 96% accuracy, above the 95% requirement, while completing processing in 45 ms, which is within the 50 ms limit and still fast enough to keep pace with the production line. Its moderate computational requirement aligns with the factory's available hardware. YOLO satisfies all three constraints at once.

### Deep Learning (generic CNN pipeline):

With 98% accuracy, this method would catch the most defects of the three, but its 75 ms processing time exceeds the 50 ms limit by a wide margin, and its high computational requirement is beyond what the hardware supports. On a real-time production line, a 25 ms overshoot per item can compound into a meaningful throughput bottleneck, and the hardware mismatch would require costly upgrades. Both factors independently rule it out.

## Decision

**YOLO is the method that satisfies all stated constraints and should be selected for the smart factory product inspection system.**

As in the vehicle case, the pattern is the same: the lightweight classical method (Traditional Image Analysis) is fast enough but not accurate enough, the heavy deep learning model is accurate enough but too slow and resource-hungry for the given hardware, and YOLO occupies the practical middle ground that meets the accuracy floor, the time ceiling, and the computation budget simultaneously. This reinforces YOLO's suitability for real-time, moderate-hardware industrial inspection tasks where both detection quality and throughput matter.

## Common Reasoning Pattern Across Both Case Studies

Both selection problems follow the same constraint-satisfaction structure: each method is evaluated against three independent gates (accuracy, timing, computation), and a method must pass all gates to be viable — passing two out of three is not sufficient. This produces a consistent trade-off spectrum across both scenarios:

- Lightweight classical methods (Viola-Jones; Traditional Image Analysis) — fastest and least resource-intensive, but accuracy falls short of the safety/quality threshold.
- Heavy deep learning pipelines — highest accuracy, but latency and computational cost exceed what real-time, moderate-hardware deployments can support.
- YOLO — positioned as a balanced, single-stage detector that meets the minimum accuracy bar while staying within timing and hardware limits, making it the constraint-satisfying choice in both the autonomous vehicle and smart factory scenarios.

This illustrates a broader principle in applied computer vision system design: the "best" method is not the one with the single highest accuracy or the single lowest latency in isolation, but the one that jointly satisfies all operational constraints of the deployment environment.